



Engine Performance Data
Cummins Inc
Columbus, Indiana 47202-3005
<http://www.cummins.com>

Automotive
X11NS6B360
FR21163

265(355)@1900
1800(1328)@1000

Configuration	CPL Code	Revision
D0R3002BX03	5356	01-Feb-2022

Compression Ratio: 17.8 :1

Displacement: 10.5 L (641 in3)

Fuel System: XPI

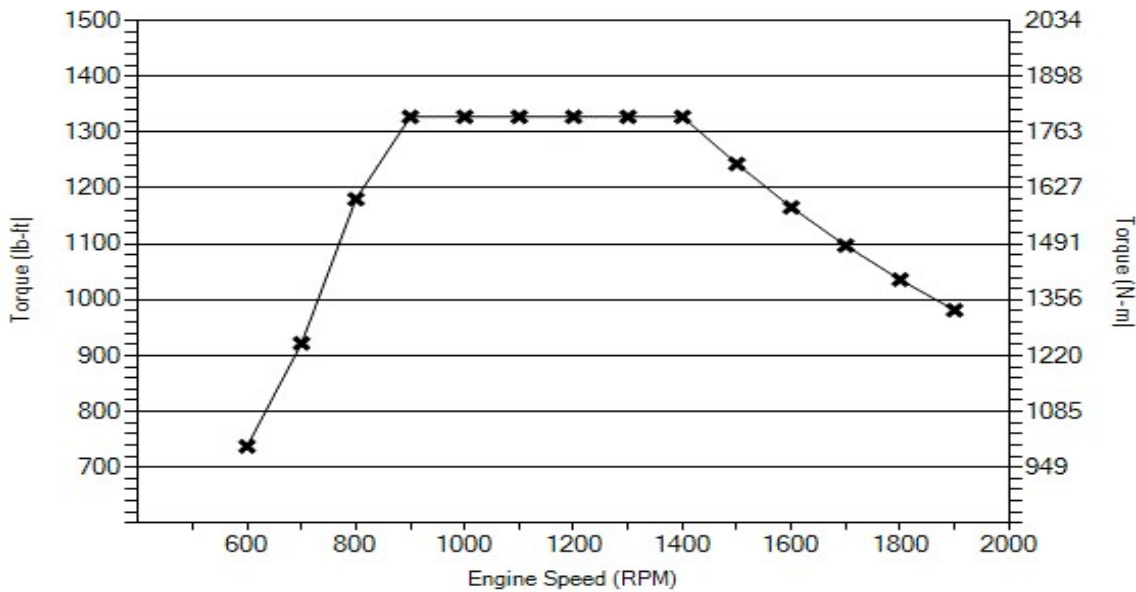
Aspiration: Turbocharged and Aftercooled

Emission Certification

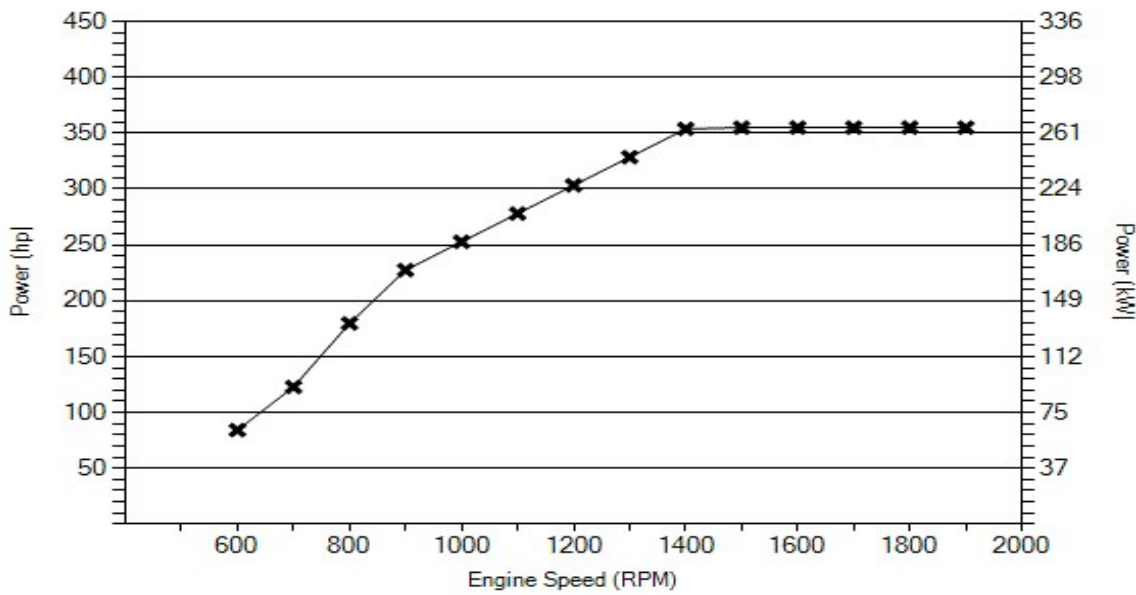
China MEP NS VIB

Rating Types

Motor Coach, Truck



Torque Output		
RPM	lb-ft	N-m
600	738	1000
700	922	1250
800	1180	1600
900	1328	1800
1000	1328	1800
1100	1328	1800
1200	1328	1800
1300	1328	1800
1400	1328	1800
1500	1243	1686
1600	1165	1580
1700	1097	1487
1800	1036	1405
1900	981	1331



Power Output		
RPM	hp	kW
600	84	63
700	123	92
800	180	134
900	227	170
1000	253	188
1100	278	207
1200	303	226
1300	329	245
1400	354	264
1500	355	265
1600	355	265
1700	355	265
1800	355	265
1900	355	265

Curves shown above represent gross engine performance capabilities obtained and corrected in accordance with SAE J1995 conditions of 100 kPa barometric pressure [91 m (300 ft) altitude], 25 deg C (77 deg F) inlet air temperature, and 1 kPa water vapor pressure with No. 2 diesel fuel.

All data based on the engine operating with fuel system, water pump, lubricating oil pump, air compressor (unloaded) and with inlet restriction and exhaust restriction at or below datasheet limits. Not included are alternator, fan, optional equipment and driven components.

Engine performance may deviate from profile with increasing elevation. Installations operating for significant periods of time at higher elevations should refer to CEB00351 for additional information regarding performance at altitude.

Tolerance within +/- 5 %

Intake Air System

Maximum allowable air temperature rise between ambient air and turbo compressor inlet	18 delta deg F	10 delta deg C
---	----------------	----------------

Cooling System

Maximum coolant temperature - engine out	216 deg F	102 deg C
Maximum coolant pressure (exclusive of pressure cap; closed thermostat at maximum no load speed)	45 psi	310 kPa
Minimum cooling capability at nominal fuel rate [Level II] with 24 km/hr (15 mile/hr) ram air speed and 50/50 ethylene glycol/water by volume		
Engine out coolant to ambient at maximum power checkpoint	109.8 delta deg F	61 delta deg C
Engine out coolant to ambient at peak torque checkpoint	109.8 delta deg F	61 delta deg C
Charge air cooler outlet to ambient at maximum power checkpoint	43.2 delta deg F	24 delta deg C
Maximum allowable pressure drop across charge air cooler and OEM CAC piping (CACDP)	3.8 in-Hg	13 kPa
Maximum coolant temperature for engine protection controls	226 deg F	108 deg C

Exhaust System

Maximum exhaust backpressure with a fully loaded DPF	14 in-Hg	46 kPa
--	----------	--------

Maximum exhaust backpressure imposed by exhaust system (at soot level after DPF regeneration or cleaning, when DPF is present)	12.4 in-Hg	42 kPa
Recommended exhaust pipe size (inner diameter)	4.6 in	116 mm
RPM for maximum exhaust backpressure	1900 RPM	1900 RPM

Fuel System

Engine fuel compatibility (consult Service Bulletin #5411406 for appropriate use of other fuels)	ULSD
--	------

Cranking System (Cold Starting Capability)

Minimum ambient temperature for unaided cold start	5 deg F	-15 deg C
Minimum ambient temperature with grid heater only	-31 deg F	-35 deg C

Performance Data

Maximum low idle speed: 800 RPM

Minimum low idle speed: 600 RPM

Nominal no load governed speed: 1925 RPM

Maximum overspeed capability: 2375 RPM

Clutch engagement torque available at the default idle speed	590 lb-ft	800 N-m
--	-----------	---------

Turbo model: HE400WG

Turbo frame size: Large

	Governed Power	Maximum Power	Peak Torque
Engine Speed	1900 RPM	1500 RPM	1000 RPM
Output Power	265 kW (355 hp)	266 kW (357 hp)	190 kW (255 hp)
Torque	1333 N-m (983 lb-ft)	1696 N-m (1251 lb-ft)	1800 N-m (1328 lb-ft)
Turbo Comp. Outlet Pressure	182.7 kPa (54.1 in-Hg)	187.5 kPa (55.5 in-Hg)	129.2 kPa (38.3 in-Hg)
Turbo Comp. Outlet Temperature	172 deg C (341 deg F)	162 deg C (323 deg F)	134 deg C (273 deg F)
Heat Rejection to Charge Air Cooler**	54 kW (3071 BTU/min)	46 kW (2633 BTU/min)	22.6 kW (1285 BTU/min)
Inlet Air Flow	371 L/s (787 ft3/min)	316 L/s (670 ft3/min)	178 L/s (377 ft3/min)
Charge (Fresh Air) Flow	25.8 kg/min (56.9 lb/min)	22 kg/min (48.5 lb/min)	12.4 kg/min (27.3 lb/min)
Exhaust Gas Flow	384 L/s (814 ft3/min)	328 L/s (695 ft3/min)	187 L/s (396 ft3/min)
Exhaust Gas Temperature	430 deg C (807 deg F)	455 deg C (852 deg F)	555 deg C (1030 deg F)
Heat Rejection to Coolant	98.3 kW (5590 BTU/min)	95.8 kW (5448 BTU/min)	84.9 kW (4828 BTU/min)
Radiator Coolant Flow*	4.3 L/s (68.2 gpm)	3.6 L/s (57.1 gpm)	2.9 L/s (46 gpm)
Fuel Consumption	53.3 kg/hr (117.5 lb/hr)	50.5 kg/hr (111.3 lb/hr)	36.9 kg/hr (81.4 lb/hr)
Brake Mean Effective Pressure	1596 kPa (231.5 psi)	2030 kPa (294.4 psi)	2173 kPa (315.2 psi)

*Radiator Coolant Flow is approximately 5% less with a continuously deaerating system.

**Heat Rejection to CAC is collected at standard engine test conditions of 25 deg C / 77 deg F turbo air inlet temperature. Please refer to the discussion section in the Advisor Installation Requirement (AIR) Report for Compressor Outlet Temperature and CAC Heat Rejection adjustment procedures for other test conditions.

***The table above represents design points to identify IQA test conditions and assist OEMs in sizing vehicle interface systems. It does not necessarily reflect advertised governed speed, maximum power or peak torque conditions.

Change Log

Date	Author	Description
11/14/2019 12:00:00 AM	Cheng Li	Update FR based on BFCEC's requirement
18/6/2020	Kerry Liu	Revision Maximum coolant temperature for engine protection controls from 113 to 108
2021/4/25	Kerry Liu	Update datasheet confidence level to final

Status for curves and data: Final-(Measured Data)

Data updated by Cheng Li

Bending moment diagrams may be available on GCE under Engine Specific Topics